

Chapter 1

Project Proposal

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1.1 Introduction and Description

Background

Multiple sclerosis is a heterogeneous, neurological autoimmune disorder wherein the immune system attacks the myelin in the central nervous system (CNS). Myelin is a component of the protective cover of axons which enable messages to be communicated around the body. When the myelin becomes damaged (demyelination), the efficiency of message passing is degraded - whilst the body may be able to repair damaged areas (lesions) to an extent (remyelination), however repeated relapses can damage oligodendrocytes, leaving them unable to produce more myelin. When the axons are left unprotected, they are more vulnerable to damage and can die. Although damage can then be somewhat mitigated by rerouting messages using undamaged nerve cells (plasticity) the damage eventually becomes too large to compensate. The disease is diagnosed with a combination of several results: the patient's health history, neurological examinations, MRI scans (the main support tool for diagnosis, as lesions will appear on scans), blood tests as well as a comparison against the McDonald criteria.

Gene expression is the process outlined by the central dogma of molecular biology wherein DNA is transcribed into RNA, and RNA is translated into functional products, and the measure of how much a gene is expressed at a given time is referred to as the gene expression level. The development of

advanced, high-throughput transcriptomic technologies like single-cell RNA sequencing (scRNA-seq) and single-nucleus RNA sequencing (snRNA-seq) allow gene expression to be captured at the individual cell levels, so the heterogeneity of cell populations can be captured.

Given that the complexity of biological networks in higher organisms arises primarily from dynamic gene interactions as opposed to the size of the genome (C-value paradox), analysing gene dynamics offers a more informative framework for understanding biological systems. Gene-gene co-expression networks are typically presented as undirected graphs where nodes are genes and the presence of edges are dependent on co-expression networks produced by calculating co-expression values based on gene expression data. These networks can be used to identify clusters of co-expressed genes (modules) and conducting guilt by association analyses.

Work to be Undertaken

1.2 Starting Point

1.3 Success Criterion

1.4 Extensions and Further Work

- Producing a multi-omic

1.5 Work Plan

1.6 Resource Declaration

I will be using my personal laptop, a 2024 Dell Inspiron 16 2-in-1, with 16GB memory and a 1TB SSD for software development. As a backup, I will use my secondary personal laptop with an external 1TB storage. I will continuously backup my code and dissertation with Git version control. I accept full responsibility for this machine and I have made contingency plans to protect myself against hardware and/or software failure.

In terms of software, I will use:

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The project also requires the use of biological datasets for training data. These datasets have been taken from <https://zenodo.org/records> which are publicly available to anyone.